Online Appendix to "The Post-ChatGPT Contraction Reaches the Crowd-Validated Margin: A Within-Tag Triple Difference and a Value-Calibrated Early-Warning System for Stack Overflow"

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This Online Appendix collects the full robustness battery, the value-distribution detail tables, the mechanism and classifier-validation detail, the descriptive counterfactual, and the literature-positioning table referenced in the main paper. Section, table, and figure numbers are prefixed "A". References to the main text (e.g. its baseline coefficient and main tables) are cross-document hyperlinks.

A.1 Full robustness battery

This appendix contains the full robustness battery. The main text retains only the consolidated Table A1 (printed once, in §5.2); the design choices and interpretation discussed in §6 are reported here in full detail.

A.1.1 Alternative AI-answerability measures

The four pre-treatment AI-answerability measures and the DDD coefficient each delivers are:

Measure	$\hat{\beta}$	SE	p	N
Structural index (baseline)	−0.108	0.046	0.021	164,351
z-score composite	−0.138	0.052	0.008	164,351
PCA composite	−0.045	0.020	0.024	164,351
Quartile rank	−0.210	0.074	0.005	164,351

The structural variant is the most conservative; the quartile rank is the largest in absolute magnitude (coarsening adds variance to the contrast).

A.1.2 Sample restrictions

Top-50 tags by pre-period volume: -0.207 ($p = 0.026$). Top-75 tags: similarly negative coefficient. Interquartile band on pre-treatment slope (75 tags closest to median slope): -0.155 ($p = 0.009$). Each of the four quartiles of pre-treatment slope returns a negative coefficient: quartile 1 (steepest decline) -0.06; quartile 2 -0.19; quartile 3 -0.08; quartile 4 (steepest growth) -0.32. The result is not driven by tags with extreme pre-trends.

Leave-one-out / jackknife: dropping the top-1 (python), top-5, top-10, and top-20 tags returns coefficients between -0.166 and -0.196. Jackknifing each of the top-10 tags individually returns a range $[-0.166, -0.136]$. The aggregate result is not an artefact of a small number of large communities.

A.1.3 Alternative outcomes

Replacing the outcome by $\log(1 + \text{unique users})$ returns $\hat{\beta} = -0.133$ ($p = 0.010$), confirming that the substitutability gradient also operates on the count of distinct authors. Counts-based estimation via PPML (Santos Silva and Tenreyro, 2006) returns $\hat{\beta} = -0.098$ ($p = 0.027$). Fractional tag counts—weighting each question by $1/n_{\text{top-100 tags}}$—return $\hat{\beta} = -0.134$ ($p = 0.005$).

A.1.4 Inference

CR1 clustering by tag: SE $= 0.052$. Two-way clustering on tag and week (Cameron et al., 2011): SE $= 0.052$ (within-tag serial correlation dominates the variance budget). Wild cluster bootstrap with 9,999 Rademacher replicates and the null imposed (Cameron et al., 2008): $p < 0.005$.

A.1.5 Continuous-substitutability specification

Replacing the binary substitutable indicator with a continuous embedding-derived substitutability score returns $\hat{\beta} = -0.039$ ($p = 0.004$) under the structural answerability measure and -0.053 ($p = 0.0009$) under the z-score composite. The continuous estimate is approximately one-third the size of the binary estimate (expected: the continuous score shrinks the substitutable vs. non-substitutable contrast).

A.1.6 Pre-trend sensitivity bounds

The sensitivity exercise of §4.2 yields the following robust 95% confidence sets under increasing $\bar{M}$ (the relative-magnitude bound on the post-period deviation from parallel trends, in units of the largest pre-period deviation):

$\bar{M}$	Robust 95% CI	Contains 0?
0.00	$[-0.098, -0.029]$	No
0.25	$[-0.117, -0.021]$	No
0.50	$[-0.136, -0.010]$	No
0.75	$[-0.159, +0.010]$	Yes (breakdown)
1.00	$[-0.186, +0.029]$	Yes
1.50	$[-0.232, +0.075]$	Yes
2.00	$[-0.281, +0.124]$	Yes

Breakdown occurs at $\bar{M} = 0.75$ under the official HonestDiD Δ^{RM} implementation with the full cluster-robust event-study VCV (§4.2). Earlier versions of this paper reported a bespoke calculation with a different threshold; the full-Σ `honestdid`-based bound is the reference value we use going forward.

A.1.7 Summary of what the robustness rules out

In combination, the robustness battery rules out:

- that the result is an artefact of one particular AI measure (all four are negative);
- that the result is driven by extreme tags (leave-one-out, IQR band, slope-quartile stratification);
- that the result is a counting artefact of multi-tag questions (fractional counts);
- that the result is a clustering or distributional artefact (two-way clustering, wild bootstrap, PPML);
- that the result is an extrapolation of pre-trends (HonestDiD Δ^{RM} full-VCV breakdown $\bar{M} = 0.75$, placebo cutoffs with opposite sign, rolling placebos);
- that the result is driven by the GitHub Copilot release (Copilot has the opposite sign);
- that the result is sensitive to specific specification choices (100% negative across 32 cells).

What the robustness does *not* rule out is the concern that the binary substitutability classification is heuristic. We address this directly in §8 with three independent classifiers and Fleiss' κ.

Table A1: Consolidated robustness battery for $\hat{\beta}_{\text{DDD}}$.

Specification	$\hat{\beta}_{\text{DDD}}$	SE	p	N
A. AI-answerability measures				
Structural index (baseline)	-0.1082**	0.0460	0.021	164,351
z-score composite	-0.1378***	0.0519	0.009	164,351
PCA composite	-0.0449**	0.0197	0.024	164,351
Quartile rank	-0.2098***	0.0722	0.005	164,351
B. Sample restrictions				
Top-50 tags by pre-period volume	-0.2067**	0.0931	0.026	85,884
Interquartile pre-slope band (75 tags)	-0.1550***	0.0578	0.009	123,185
Slope quartile 4 (steepest growth)	-0.3221**	0.1260	0.017	40,775
Slope quartile 1 (steepest decline)	-0.0554	0.0914	0.550	41,185
Leave-one-out: drop top-1 (python)	-0.1655***	0.0533	0.002	162,526
Leave-out: drop top-5	-0.1926***	0.0546	0.001	155,278
Leave-out: drop top-10	-0.1957***	0.0563	0.001	146,506
Leave-out: drop top-20	-0.1842***	0.0673	0.008	129,152
C. Outcome and method alternatives				
Outcome log(1 + unique users)	-0.1329***	0.0515	0.010	164,351
PPML (Poisson FE)	-0.0984**	0.0444	0.027	164,351
Two-way clustering (tag + week)	-0.1378***	0.0522	0.008	—
Fractional tag counts (Q/n_{tags})	-0.1345***	0.0471	0.005	164,968
D. Identification: placebos, Pre-trend sensitivity, co-shock				
Placebo cutoff 2020-05-30 (pre-only)	0.0625***	0.0235	0.009	99,185
Placebo cutoff 2020-11-30 (pre-only)	0.0643***	0.0198	0.002	99,185
Placebo cutoff 2021-05-30 (pre-only)	0.0670***	0.0200	0.001	99,185
Placebo cutoff 2021-11-30 (pre-only)	0.0620***	0.0226	0.007	99,185
Placebo cutoff 2022-05-30 (pre-only)	0.0592**	0.0231	0.012	99,185
Real cutoff (2022-11-30)	**-0.1378*****	**0.0519**	**0.009**	**164,351**
Copilot placebo (2022-06-21, pre-ChatGPT)	0.0575**	0.0222	0.011	99,185
Dual-shock: AI×Sub×Copilot	0.0578**	0.0226	0.012	164,351
Dual-shock: AI×Sub×ChatGPT	-0.1573***	0.0434	0.000	164,351
Rolling placebo mean (29 cutoffs)	0.0574	0.0068	—	—
share of rolling placebos < -0.10	0%			
HonestDiD Δ^{RM} full-VCV, $\bar{M} = 0.0$, robust CI $[-0.098, -0.029]$	excludes 0			
HonestDiD Δ^{RM} full-VCV, $\bar{M} = 0.50$, robust CI $[-0.136, -0.010]$	excludes 0			
HonestDiD Δ^{RM} full-VCV, $\bar{M} = 0.75$, robust CI $[-0.159, +0.010]$	contains 0 (breakdown)			
HonestDiD Δ^{RM} full-VCV, $\bar{M} = 1.0$, robust CI $[-0.186, +0.029]$	contains 0			
HonestDiD Δ^{RM} full-VCV, $\bar{M} = 1.5$, robust CI $[-0.232, +0.075]$	contains 0			
HonestDiD Δ^{RM} full-VCV, $\bar{M} = 2.0$, robust CI $[-0.281, +0.124]$	contains 0			
E. Wild cluster bootstrap				
Wild bootstrap (Rademacher, 9,999 reps, cluster=tag)	-0.1378	0.0519	$p_{\text{boot}} < 0.0001$	—
F. Cluster-robust inference (CR3 jackknife)				
CR3 cluster-robust, $n_{\text{tag}} = 100$	-0.1082	0.0495	0.031	164,351

Notes. Baseline regression: $\log(1+Q_{t,w,k}) = \beta_{\text{DDD}}\,(\text{AI}_t{\cdot}\text{Post}_w{\cdot}\text{Sub}_k) + \gamma_1(\text{AI}_t{\cdot}\text{Post}_w) + \gamma_2(\text{AI}_t{\cdot}\text{Sub}_k) + \gamma_3(\text{Post}_w{\cdot}\text{Sub}_k) + \alpha_{t,k} + \delta_w + \varepsilon$. FE: tag × question-type and week. Errors clustered by tag (CR1). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Specification curve in Figure A1 reports 32 alternative specifications; 100% are negative and significant at $p < 0.05$.

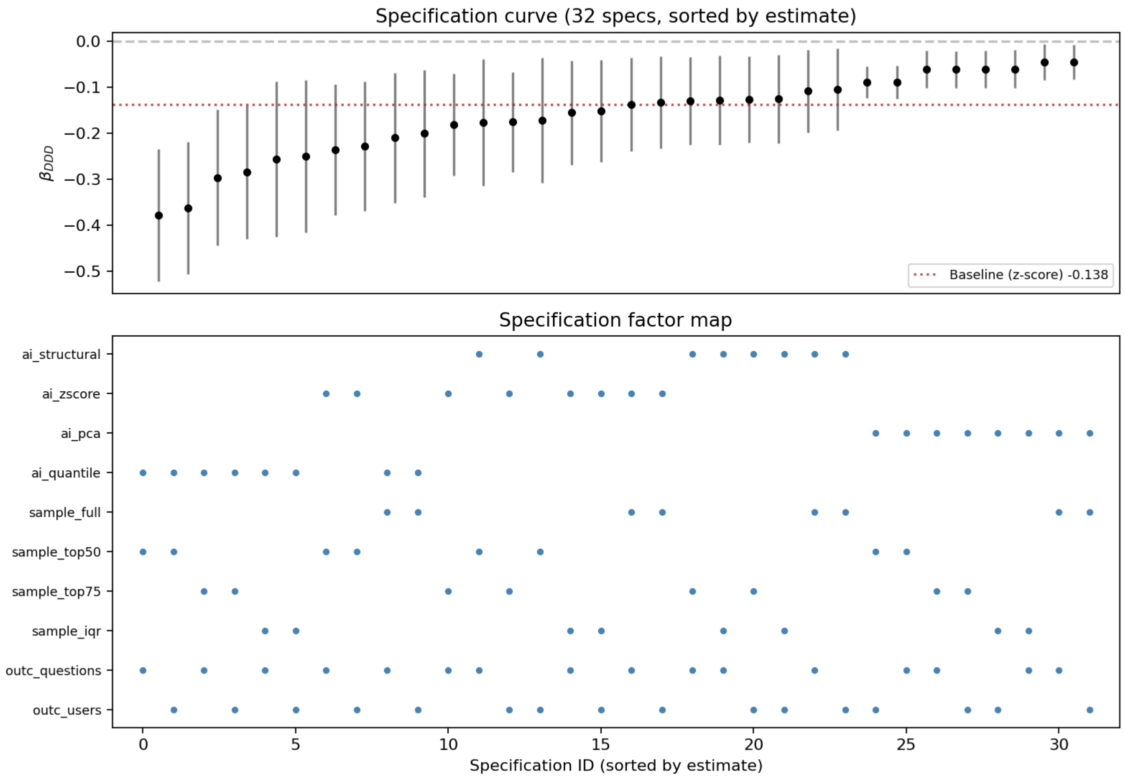

Figure A1: Specification curve over the full 32-cell factorial grid. All coefficients negative; all significant at $p < 0.05$.

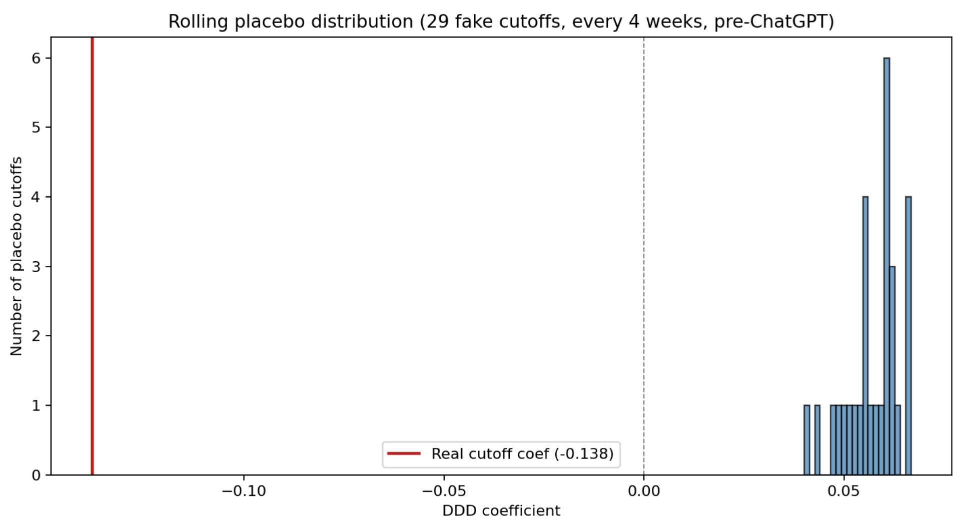

Figure A2: Rolling-placebo distribution. 29 placebo cutoffs every four weeks across 2020-07-01 to 2022-09-01. Vertical line: real cutoff coefficient -0.138.

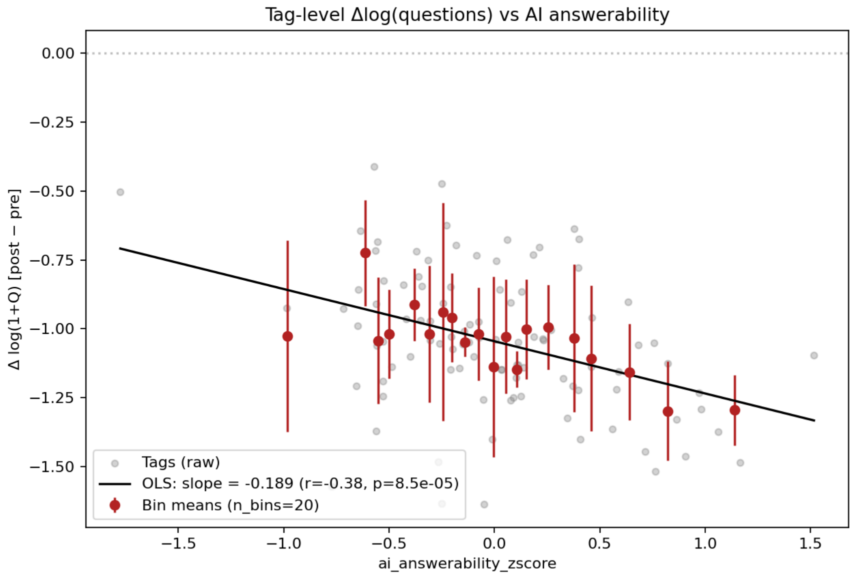

Figure A3: Tag-level $\Delta \log(1 + Q)$ (post minus pre) against pre-treatment AI answerability ($n = 100$ tags).

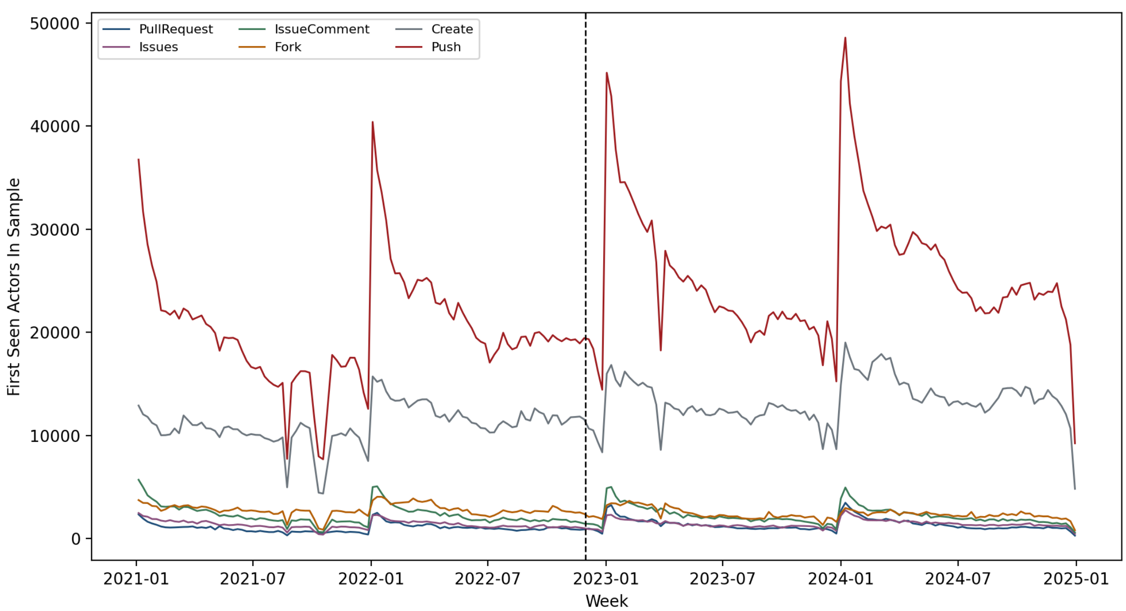

Figure A4: First-seen GitHub actors by event type in the entry-oriented GH Archive sample, 2021–2024.

A.2 Value-distribution detail

This appendix reports the per-bin, threshold, and robustness detail behind the value-margin result of the main text (§5.4).

Table A2: Mutually exclusive score-bin effects for accepted, non-closed posts. Each row refits the baseline DDD with the dependent variable equal to the count of posts in the score bin. Unlike cumulative threshold outcomes, these bins show where in the crowd-score distribution the negative association is concentrated. The MDE column reports the approximate two-sided 5% test, 80% power minimum detectable effect, computed as $2.80 \times SE$.

Score bin	$\hat{\beta}_{DDD}$	SE	MDE_{80}	p	95% CI	N obs	N posts
score = 0	-0.2080	0.0469	0.131	0.0000	[-0.301, -0.115]	164,351	1,620,830
score = 1	-0.2029	0.0440	0.123	0.0000	[-0.290, -0.116]	164,351	895,360
score = 2	-0.2121	0.0475	0.133	0.0000	[-0.306, -0.118]	164,351	341,274
score = 3	-0.1641	0.0429	0.120	0.0002	[-0.249, -0.079]	164,351	137,393
score = 4	-0.1240	0.0413	0.116	0.0034	[-0.206, -0.042]	164,351	64,650
score $\geq$ 5	-0.0906	0.0571	0.160	0.1155	[-0.204, 0.023]	164,351	128,126

Table A3: Score-threshold sensitivity of the high-value curated artefact DDD. Each row re-fits the baseline triple difference with the dependent variable being the count of curated artefacts (accepted, non-closed) whose net score meets the threshold. The trajectory of the coefficient as k grows is the relevant object: a flat trajectory indicates that the displacement is proportional across the value distribution; an attenuating trajectory indicates that the effect concentrates on lower-value artefacts.

Threshold	$\hat{\beta}_{DDD}$	SE	p	95% CI	N obs	N artefacts
score $\geq$ 0	-0.1183	0.0487	0.0169	[-0.215, -0.022]	164,351	3,187,633
score $\geq$ 1	-0.1308	0.0505	0.0110	[-0.231, -0.031]	164,351	1,566,803
score $\geq$ 2	-0.1446	0.0553	0.0103	[-0.254, -0.035]	164,351	671,443
score $\geq$ 3	-0.1255	0.0577	0.0319	[-0.240, -0.011]	164,351	330,169
score $\geq$ 4	-0.1110	0.0596	0.0655	[-0.229, 0.007]	164,351	192,776
score $\geq$ 5	-0.0906	0.0571	0.1155	[-0.204, 0.023]	164,351	128,126

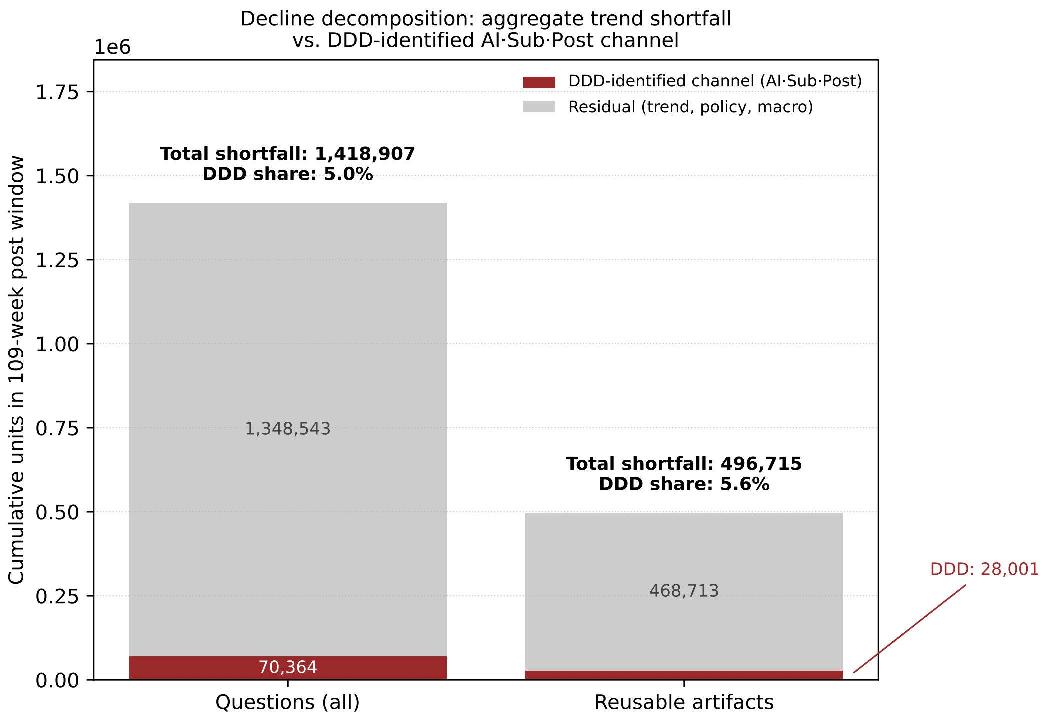

Figure A5: Decomposition of the post-ChatGPT shortfall. Dark bar (top): residual outside the DDD-estimated channel. Red bar (bottom): DDD-estimated channel. The DDD-estimated shortfall is negative and statistically significant under the maintained assumptions; the residual is a descriptive benchmark, not an alternative causal estimate, and its size simply reflects how much of the aggregate decline the within-tag DDD does not estimate.

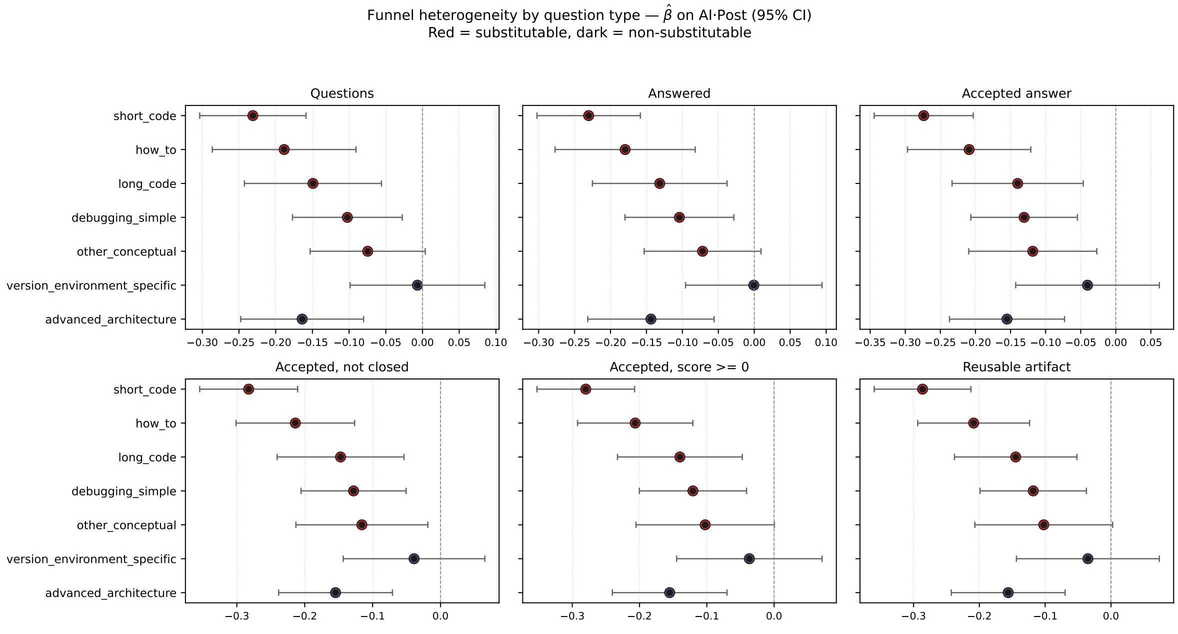

Figure A6: Funnel heterogeneity by question type. Each panel reports $\hat{\beta}_k$ on AI$\cdot$Post within question type k (tag and week fixed effects, tag-clustered CIs). Red: substitutable; dark: non-substitutable. The version-environment-specific category is null across all funnel stages.

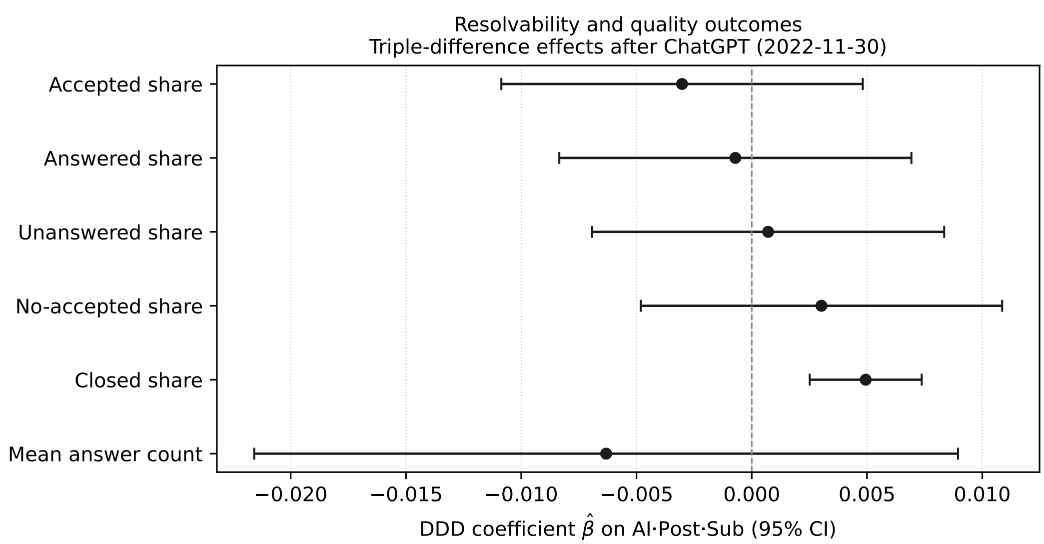

Figure A7: DDD coefficients on resolvability and quality shares at the (tag, week, question-type) cell. Weighted OLS with (tag $\times$ question type) and week fixed effects; tag-clustered 95% CIs. Five of six estimates are statistically indistinguishable from zero; closed share rises by +0.005 ($p < 0.001$).

Table A4: Romano–Wolf step-down family-wise error-rate correction for the six cumulative score-threshold tests, computed by a cluster bootstrap resampling tags (B=999 replications, recentred studentized statistics). The family-wise adjustment exploits the strong positive dependence across the nested thresholds. The low thresholds remain significant after correction; the elite tail does not.

Threshold	$\hat{\beta}_{\mathrm{DDD}}$	SE	p_{single}	p_{RW}
score ≥ 0	-0.1183	0.0487	0.027	0.077
score ≥ 1	-0.1308	0.0505	0.033	0.068
score ≥ 2	-0.1446	0.0553	0.028	0.066
score ≥ 3	-0.1255	0.0577	0.058	0.093
score ≥ 4	-0.1111	0.0596	0.108	0.125
score ≥ 5	-0.0906	0.0571	0.169	0.169

Table A5: Voting-baseline-normalised high-value curated artefact DDD. The post-period score is rescaled by the ratio of pre-period to post-period mean absolute score at the tag level, addressing the concern that post-ChatGPT voting volume falls and mechanically depresses scores. Rescaling factors are clipped to [0.5, 3.0] to avoid extrapolation in low-traffic tags.

Threshold	$\hat{\beta}_{DDD}$	SE	p	95% CI	N artefacts
score ≥ 1 (voting-normalised)	-0.1438	0.0542	0.0094	[-0.251, -0.036]	1,555,212
score ≥ 2 (voting-normalised)	-0.1793	0.0494	0.0005	[-0.277, -0.081]	671,673
score ≥ 5 (voting-normalised)	-0.0979	0.0427	0.0241	[-0.183, -0.013]	134,838

A.3 Mechanism detail: resolvability and survivor composition

A.4 Classifier and treatment-measurement detail

A.5 Descriptive counterfactual: linear trend extrapolation

The main text reports the AI-channel-attributable shortfall (∼70,000 missing questions) using the DDD coefficient applied cell by cell. For context, we describe in this appendix the relationship between this DDD-implied counterfactual and a simpler descriptive counterfactual based on linear extrapolation of the pre-ChatGPT aggregate trend.

Table A6: Sensitivity of the voting-baseline-rescaled high-value DDD to the clip range applied to the per-tag pre/post mean-absolute-score ratio. The four clip ranges return *numerically identical* estimates because every tag's ratio already lies within the tightest band $[0.7, 2.0]$, so the clip never binds: the rescaling is applied (these coefficients differ from the unrescaled -0.131, -0.145, -0.091) but its result does not depend on the clip choice at all. In particular the elite-tail estimate (score ≥ 5) is significant even with no clipping, which addresses the concern that the clip is a researcher degree of freedom that rescues the result.

Clip range	score ≥ 1	score ≥ 2	score ≥ 5
No clip	-0.144^*	-0.179^*	-0.098^*
$[0.3, 5.0]$	-0.144^*	-0.179^*	-0.098^*
$[0.5, 3.0]$ (main)	-0.144^*	-0.179^*	-0.098^*
$[0.7, 2.0]$	-0.144^*	-0.179^*	-0.098^*

* $p < 0.05$ (tag-clustered). All tag ratios lie within $[0.7, 2.0]$, hence the rows coincide.

A.5.1 Linear extrapolation of the pre-ChatGPT trend

The pre-ChatGPT log-linear regression of the weekly total on time is $\log Q_w = c + \rho w$ with $\hat{\rho} = -0.00333$ per week and $R^2 = 0.80$. Extrapolating into the post-ChatGPT window implies a counterfactual aggregate of $\sim$3.09 million questions; observed is $\sim$1.67 million; the implied break from trend is $\sim$1.42 million questions or -46% from the trend-implied path. Figure A8 shows the two trajectories with the shaded shortfall area.

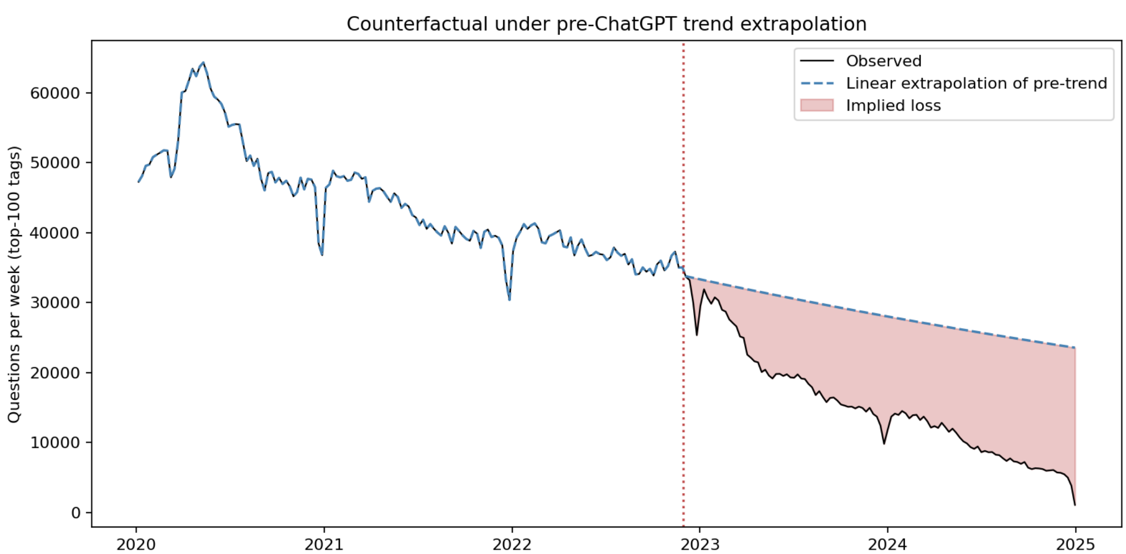

Figure A8: Aggregate weekly questions observed vs. linear extrapolation of pre-ChatGPT trend. Vertical line: ChatGPT release. Shaded area: implied break from trend (descriptive only, not causal).

Table A7: Heterogeneity of the funnel response by question type. Each cell reports $\hat{\beta}$ on AI·Post from a tag-clustered DiD fitted within that question type only (since Sub_k is constant within a type). Standard errors in parentheses below each coefficient. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. † Non-substitutable question type (control group in main DDD).

Question type	Questions	Answered	Accepted answer	Accepted, not closed	Accepted, score ≥ 0	Funnel-qualified post
short_code	−0.231***	−0.230***	−0.274***	−0.283***	−0.280***	−0.286***
	(0.037)	(0.036)	(0.036)	(0.036)	(0.037)	(0.037)
how_to	−0.189***	−0.180***	−0.209***	−0.214***	−0.206***	−0.209***
	(0.049)	(0.049)	(0.044)	(0.044)	(0.043)	(0.043)
long_code	−0.149**	−0.132**	−0.140**	−0.147**	−0.140**	−0.145**
	(0.047)	(0.047)	(0.047)	(0.047)	(0.047)	(0.047)
debugging_simple	−0.102**	−0.104**	−0.131***	−0.128**	−0.121**	−0.118**
	(0.038)	(0.038)	(0.038)	(0.039)	(0.040)	(0.041)
other_conceptual	−0.075	−0.072	−0.118*	−0.116*	−0.102	−0.102
	(0.040)	(0.041)	(0.046)	(0.049)	(0.052)	(0.053)
version_environment†	−0.007	−0.001	−0.040	−0.039	−0.037	−0.035
	(0.046)	(0.048)	(0.052)	(0.053)	(0.054)	(0.055)
advanced_architecture†	−0.164***	−0.144**	−0.155***	−0.155***	−0.155***	−0.156***
	(0.042)	(0.044)	(0.041)	(0.042)	(0.043)	(0.044)

A.5.2 Why this descriptive number is not the causal effect

The 1.42-million figure is *descriptive only*. It attributes the entire break from the pre-trend to the ChatGPT release; this attribution is not justified by the data, because the pre-treatment trend itself reflects a mix of forces (maturation of the platform, secular shift to private channels, prior LLM access via Codex and earlier GPT releases). The DDD design corrects for this by identifying a within-tag, within-question-type residual: that residual corresponds to roughly ∼70,000 missing questions, not 1.42 million.

We retain the descriptive number in the appendix for two reasons: (a) it provides a maximum-credible upper bound on the size of the AI-mediated channel that the DDD is narrowing within; and (b) it makes visible how much of the post-ChatGPT decline is *not* identified by our design (approximately 96%). The main paper uses the DDD-implied counterfactual (∼70,000) and explicitly states the DDD's identification scope.

A.6 Positioning against recent literature

References

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427,

Table A8: Decomposition of the post-ChatGPT decline in Stack Overflow into (i) the aggregate trend-implied shortfall and (ii) the DDD-estimated AI·Sub·Post component. The DDD component quantifies only the conditional association estimated by the triple difference under its maintained assumptions. The residual reflects all other co-occurring sources of decline (secular trends, platform policies, macro shocks).

	Questions (all)	Funnel-qualified posts
Pre weeks	152	152
Post weeks	109	109
Pre weekly avg	44,159	17,411
Post weekly avg	15,332	4,965
Observed post total	1,671,197	541,162
Trend-implied CF total	3,090,104	1,037,877
Trend-implied shortfall	1,418,907	496,715
DDD-estimated shortfall	70,364	28,001
DDD / shortfall share	5.0%	5.6%
Residual after DDD	1,348,543	468,713

2008.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29(2):238–249, 2011.

Martin Quinn and Dominik Gutt. Heterogeneous effects of generative AI on knowledge seeking in online communities. *Working paper*, 2024. SSRN working paper 4522386.

J. M. C. Santos Silva and Silvana Tenreyro. The log of gravity. *Review of Economics and Statistics*, 88(4):641–658, 2006.

Table A9: Triple-difference (DDD) effects on resolvability and quality outcomes at the tag-week-question-type cell. All regressions are weighted by cell question count and include tag×qtype and week fixed effects; standard errors clustered at the tag level.

Outcome	$\hat{\beta}_{DDD}$	SE	p	95% CI	N obs
Accepted share	-0.0030	(0.0040)	0.4460	[-0.0109, 0.0048]	164,351
Answered share	-0.0007	(0.0038)	0.8539	[-0.0083, 0.0069]	164,351
Unanswered share	0.0007	(0.0038)	0.8539	[-0.0069, 0.0083]	164,351
No-accepted share	0.0030	(0.0040)	0.4460	[-0.0048, 0.0109]	164,351
Closed share	0.0049	(0.0012)	0.0001	[0.0025, 0.0074]	164,351
Mean answer count	-0.0063	(0.0077)	0.4135	[-0.0216, 0.0089]	164,351

Table A10: Pre/post means by AI-answerability group (median split on the AI structural index). Means are weighted by cell question count. Δ = Post − Pre; DiD = HighAI Δ − LowAI Δ.

	LowAI			HighAI			DiD
Variable	Pre	Post	Δ	Pre	Post	Δ	
Body length (chars)	2,060	2,398	+339	1,916	2,300	+384	+45
Accepted share	0.394	0.322	−0.072	0.473	0.396	−0.077	−0.005
Answered share	0.780	0.725	−0.055	0.832	0.781	−0.051	+0.005
Mean answer count	1.082	0.939	−0.143	1.241	1.081	−0.159	−0.016
Mean score	0.980	0.588	−0.392	0.691	0.478	−0.214	+0.178
Closed share	0.036	0.035	−0.001	0.062	0.060	−0.002	−0.001
Code share	0.822	0.825	+0.002	0.899	0.896	−0.003	−0.006

Table A11: Embedding-based external validation of the AI-answerability proxy. For each tag we compute the mean cosine-similarity differential of a random sample of pre-ChatGPT questions against two sets of exemplar prompts: six that a frontier LLM answers well (short syntactic and how-to questions) and six that require private or organisation-specific context that a frontier LLM cannot reproduce. The mean differential is then correlated with the four pre-treatment AI-answerability index variants. Positive correlation indicates that the historical-feature-based index is consistent with an independent embedding-based proxy. Embeddings: `sentence-transformers all-MiniLM-L6-v2`, $N = 1000$ pre-ChatGPT questions.

Structural index variant	Pearson r	p	Spearman ρ	p
Structural	0.409	0.0000	0.509	0.0000
Zscore	0.444	0.0000	0.502	0.0000
Pca	0.478	0.0000	0.460	0.0000
Quantile	0.504	0.0000	0.525	0.0000

Table A12: Partial-regression test for the embedding-based validation of the AI-answerability proxy. Each row reports the OLS coefficient of the embedding-based score on the `ai_answerability_structural` proxy, adding tag-level controls progressively: log(1 + pre-period questions), tag maturity (weeks since first appearance), and pre-ChatGPT accepted-answer rate. A positive coefficient that survives the addition of popularity-, maturity-, and accepted-rate controls indicates that the embedding-based validation does not merely capture historical popularity. Standard errors are HC1-robust.

Specification	Controls	Coef.	SE	t	p	R^2
(1) bivariate	none	0.0474	0.0160	2.97	0.0030	0.167
(2) + log(vol)	`log_pre_volume`	0.0549	0.0181	3.03	0.0024	0.183
(3) + maturity	+ `tag_maturity_weeks_pre`	0.0731	0.0109	6.71	$< 10^{-10}$	0.291
(4) + accepted rate	+ `accepted_answer_rate_pre` (component of proxy)	0.0259	0.0362	0.71	0.4751	0.310

Table A13: Revealed-answerability validation of the AI-answerability proxy. The validation outcome is built only from pre-ChatGPT Stack Overflow resolution signals: accepted answers, answered-and-not-closed status, fast first answer, and accepted fast answer. Positive values indicate that tags with higher AI-answerability also had objectively more resolvable pre-treatment questions.

Validation outcome	Metric	Coef.	N
Tag-level revealed score, Pearson	pearson	0.563	100
Tag-level revealed score, Spearman	spearman	0.630	100
Tag-level high-answerability share	pearson	0.649	100
Fast answer within 24h share	pearson	0.568	100
Accepted fast answer within 24h share	pearson	0.498	100
Answered and not closed share	pearson	0.270	100
Question-level revealed score, Spearman	spearman	0.138	30,241
AUC predicting high revealed answerability	AUC	0.564	30,241

Table A14: Question-level discrimination of the tag-level answerability proxy for revealed high answerability, overall and by pre-treatment answerability quartile. The proxy is a coarse, tag-level instrument and is weak at the individual-question level (AUC near 0.5–0.57 throughout); we report the stratified values rather than the pooled AUC alone so the reader can see that the instrument carries tag-level, not question-level, signal.

Stratum	AUC	N
Overall	0.566	30,241
Answerability quartile 1	0.589	7,716
Answerability quartile 2	0.470	8,599
Answerability quartile 3	0.517	9,725
Answerability quartile 4	0.500	4,201

Table A15: Positioning of this paper relative to recent empirical work on the post-ChatGPT shock to programming Q&A platforms. A check mark indicates that the cited study reports a substantive empirical result on the corresponding margin. "Task-level DDD" refers to estimation via a triple difference exploiting variation in task-level LLM substitutability; "support-post funnel" refers to decomposing the effect through stages: questions → answered → accepted → accepted-not-closed → accepted-non-negative-score → funnel-qualified post.

Study	Quantity decline	Quality / composition	Users / retention	Answers	Novelty / readability	Task-level DDD	Support-post funnel
del Rio-Chanona et al. (2024)	✓	✓					
Burtch et al. (2024)	✓	✓	✓				
Xue et al. (2026, ISR)	✓	✓	✓	✓			
Shan & Qiu (2025, ISR)	✓	✓		✓			
Quinn and Gutt (2024)		✓			✓		
This paper	✓	✓				✓	✓